01 DECISION SNAPSHOT

M1 evidence checkpoint

Prepared for Wei He | 17 October 2026 | Synthetic PV investigation world

M1 ACCEPTANCE PENDING - CORE EVIDENCE ONLY

386	4.92 s	0
Passed cases in supplied JUnit	Actual partial trace duration	M1 RGB frames in core trace

Recorded core checkpoint - source hashes match at report build

The verified recording contains 2,461 actual MuJoCo/PV/thermal states through tick 2,460. It ends after a partial action sequence; a finalized recording file does not mean a completed pilot or milestone. [S1-S4]

The supplied test report records 386 passed, 0 failed, 0 errors and 0 skipped cases. Counts come from that single JUnit file; overlapping component and reviewer selections are not added. [S5]

Complete development pilots supplied: 0. Captured M1 RGB frames in the core trace: 0.

Scope of this report

Established by core inputs	Required separate evidence
Finite actual motion and instrument timing; explicit surrogate output and heat accounting.	Real moving-camera images, three completed development pilots and rendered replay.
Recorded actions replayed without calling a policy; injected accounting fixture retained.	Independent full M1 review and Wei's decision before any M2 work.

This is a private evaluator report under trusted-policy/local. It makes no physical validation, diagnosis-performance or hostile-code isolation claim.

02 EVIDENCE AND READINESS

What passed, what remains

Current report inputs are explicit; historical checkpoints remain labelled.

Check	Evidence state	Interpretation
Supplied automated suite	386 / 386 passed	JUnit file; current-code scope depends on its own run provenance.
Core replay	Passed	Matched source recording; exact tick checks plus private state and runtime metrics.
Independent review	Blocked pending desktop	Review verdict applies only to its stated scope.
New M1 RGB in core trace	0	No image can be inferred from numeric state or a scientific geometry plot.
Desktop batch	Pending	Three seed-specific pilot/replay pairs are required for the planned gate.
Full milestone gate	Pending	M2 remains outside this report's authorization.

Recorded action chronology

Attempt at (s)	Action ID	Recorded request
0.020	1	move_camera
1.910	1	move_camera
1.910	2	set_voltage
2.110	2	set_voltage
2.110	3	probe_voltage
4.620	4	wait

Repeated IDs remain visible because accepted retries and conflicts are part of the authoritative trace. This table lists requests, not inferred action success.

03 ELECTRICAL AND THERMAL STATE

Energy through the partial trace

Saved state at each 2 ms tick; plotting does not rerun the simulator. [S2-S4]

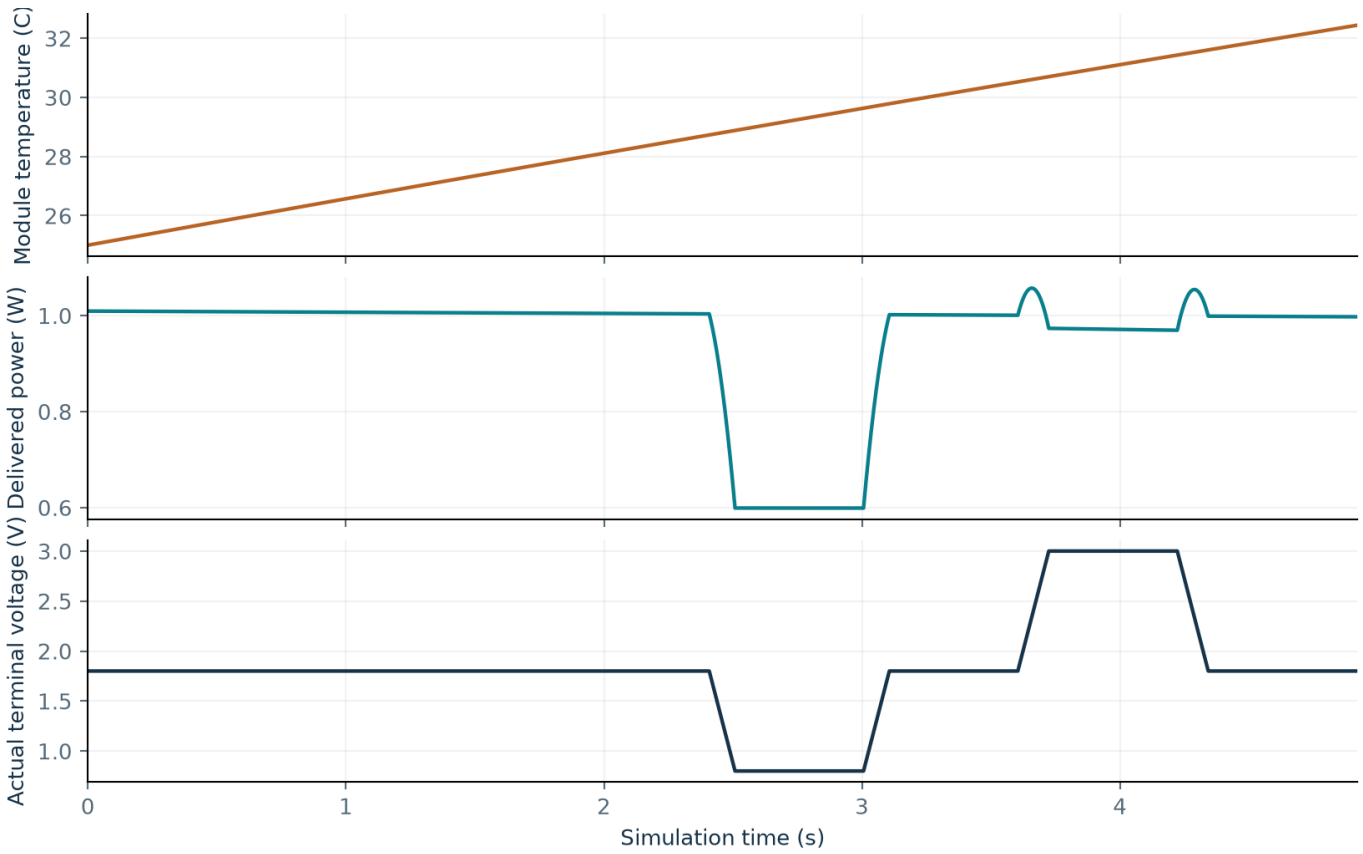


Figure 1. Temperature rises from 25.00 to 32.44 C while load/probe operations change delivered power. These are synthetic model states, not calibrated hardware measurements.

Recorded energy account	Joules
Absorbed incident energy	239.112000
Delivered electrical output	4.694841
Integrated heat loss	11.162274
Thermal storage change	223.254886

Residual = absorbed - output - heat loss - storage = -1.492e-11 J. Output integration freezes each tick's electrical input; the thermal step integrates linear cooling exactly. Plotted electrical values are end-of-tick samples, so integrating the plotted curve is not the ledger rule.

Across 2,461 true samples, max |P - V x I| = 0.000e+00 W.

04 ACTUAL MOTION

Servos follow finite trajectories

Saved MuJoCo positions; dashed curves are saved finite servo setpoints. [S2]

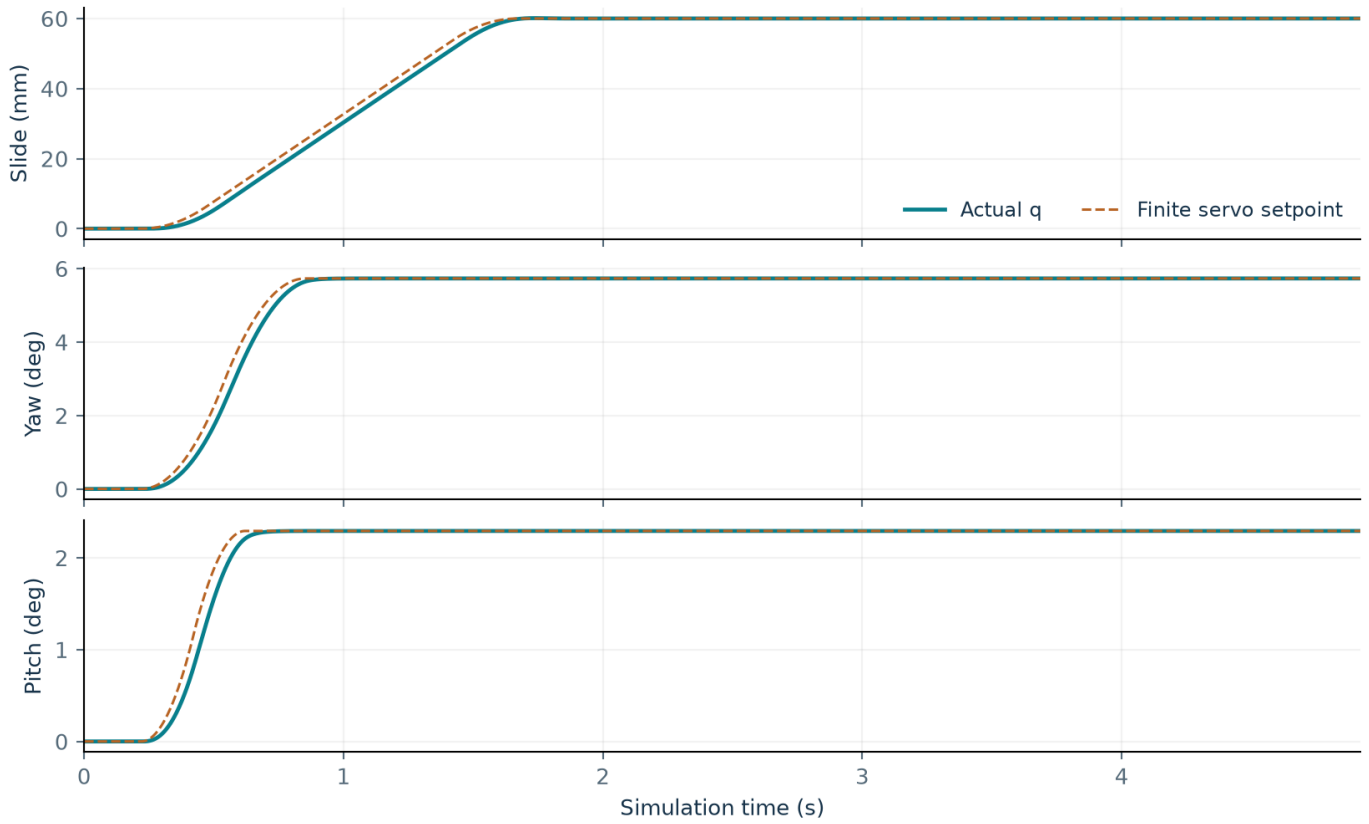


Figure 2. Slide, yaw and pitch evolve over simulation time. A setpoint curve is a command trajectory, while the solid curve is the achieved joint coordinate. Endpoint agreement alone does not establish settlement throughout every action.

Coordinate	Final actual	Final setpoint	Absolute error
Slide (mm)	60.000000	60.000000	3.469e-14
Yaw (deg)	5.729578	5.729578	7.156e-15
Pitch (deg)	2.291831	2.291831	3.976e-15

Recorded travel: slide 0.060219 m, yaw 0.100000 rad, pitch 0.040000 rad. The actuator-work proxy is 0.000427332 J. It is reported separately and is not deducted from PV output.

05 SCENE AND CAMERA

Recorded camera path in 3D

Scientific geometry figure from configuration and actual pose records. [S2, S4]

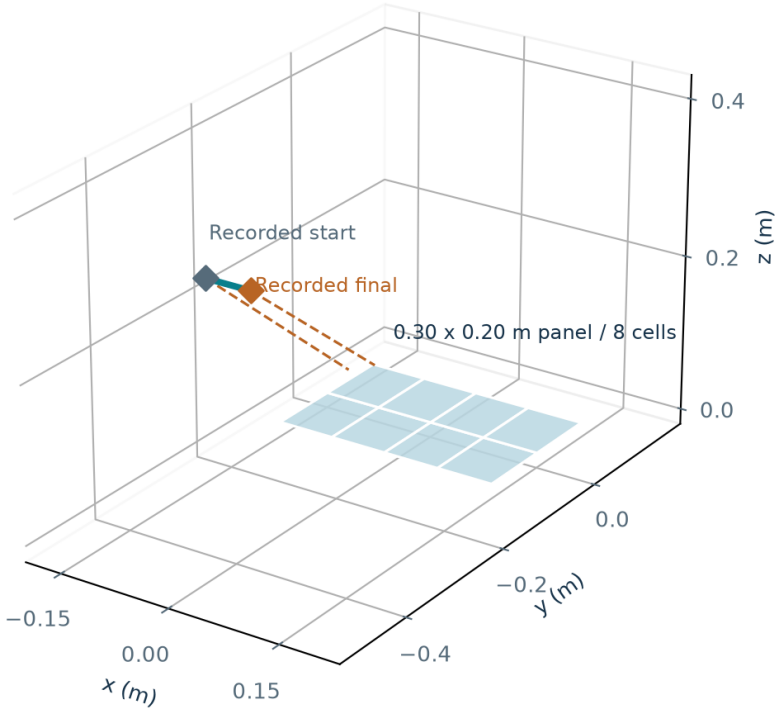


Figure 3. Panel geometry comes from the recorded configuration; camera centres and orientation rays come from actual stored transforms. The drawing omits the rig housing, lighting and occluders. It is not a MuJoCo RGB image or a visibility test.

Recorded camera centre	x, y, z (m)
Start	0.000, -0.450, 0.350
Final	0.060, -0.450, 0.350

M1 RGB not yet available

The core recording has no RGB frames. No placeholder image has been generated.

06 PROVENANCE AND REVIEW

A traceable checkpoint

Source snapshots, figure measurements and hashes accompany this PDF.

Independent review

Supplied verdict: **BLOCKED_PENDING_DESKTOP**. The review file is preserved without edits in the source snapshots. Its verdict is not promoted beyond the scope stated in that file. [S6]

Matched replay report: passed; 2,461 private truth records and 49 numeric metric values compared. Maximum private metric difference: 0.0. [S7] Injected replay-accounting value: 1.25 s. This is a known test fixture, not measured policy runtime. No benchmark inference latency was measured. [S8]

Integrity and version scope

Manifest and every listed public/private artifact passed SHA-256 verification before plotting. The source checkpoint was completed at 2026-10-17T14:27:05.550558+00:00. At this report build, 0 source/configuration paths differ from that checkpoint. Historical traces remain valid evidence of their recorded version; they do not establish successful execution of later code.

Source	Retained input	SHA-256 prefix
S1	Verified source recording manifest	b616608c58e8cb63
S2	Private actual-state trace	c885b14a759bb86b
S3	Recorded energy and cost metrics	e18a90fd0f988711
S4	Expanded private configuration	fef54964e97a8c39
S5	JUnit test evidence	3aeb5cadb1f593fb
S6	Independent review input	d273ece5209ac5a0
S7	Matched recorded-action replay report	da11cfe1cddf2239
S8	Independent core replay fixture source	5899ce06e6ab9927

Reproduction boundary

The report generator reads artifacts only. It opens no graphics context, executes no policy and runs no new episode. Full hashes, input locations, preserved snapshots, figure hashes, measurements, runtime package versions and the generation command are in the adjacent provenance JSON. No M2 authorization is inferred from this report.